USING SQL

DATA ANALYSIS

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This case study uses MySQL. To successfully answer the questions, I have used the following concepts of SQL:

- Basic aggregation
- Date time functions
- Views
- CTEs
- Windows function
  a. Ranking
  b. Aggregate
  c. Value-based
- Rounding functions (round,ceil,floor)

Questions

- Q1. Find out the total profit, sales, discount and shipping cost according to the segment.
- Q2. Top 10 corporate customers who gave the best sales.
- Q3. How many order received in India on each segment?
- Q4. What would be the % of profit with respect to sales for santiago state by each order priority.
- Q5. What is the avg shipping cost of 1st class for top 15 customers in south-east asia for home office site?
- Q6. What are the total no of orders for each category by each segment order date wise?
- Q7. How many product shipped by first class in each priority.
- Q8. Find out total sales and profit for each category with low order priority and zero discount.
- Q9. Name the cities of Australia with % of sales?
- Q10. What is the total sales and quantity for top 5 samsung products in APAC Market for the year 2014?
- Q11. What % of shipping cost contribute in sales for Nokia Products in NEW ZEALAND country ?

Questions

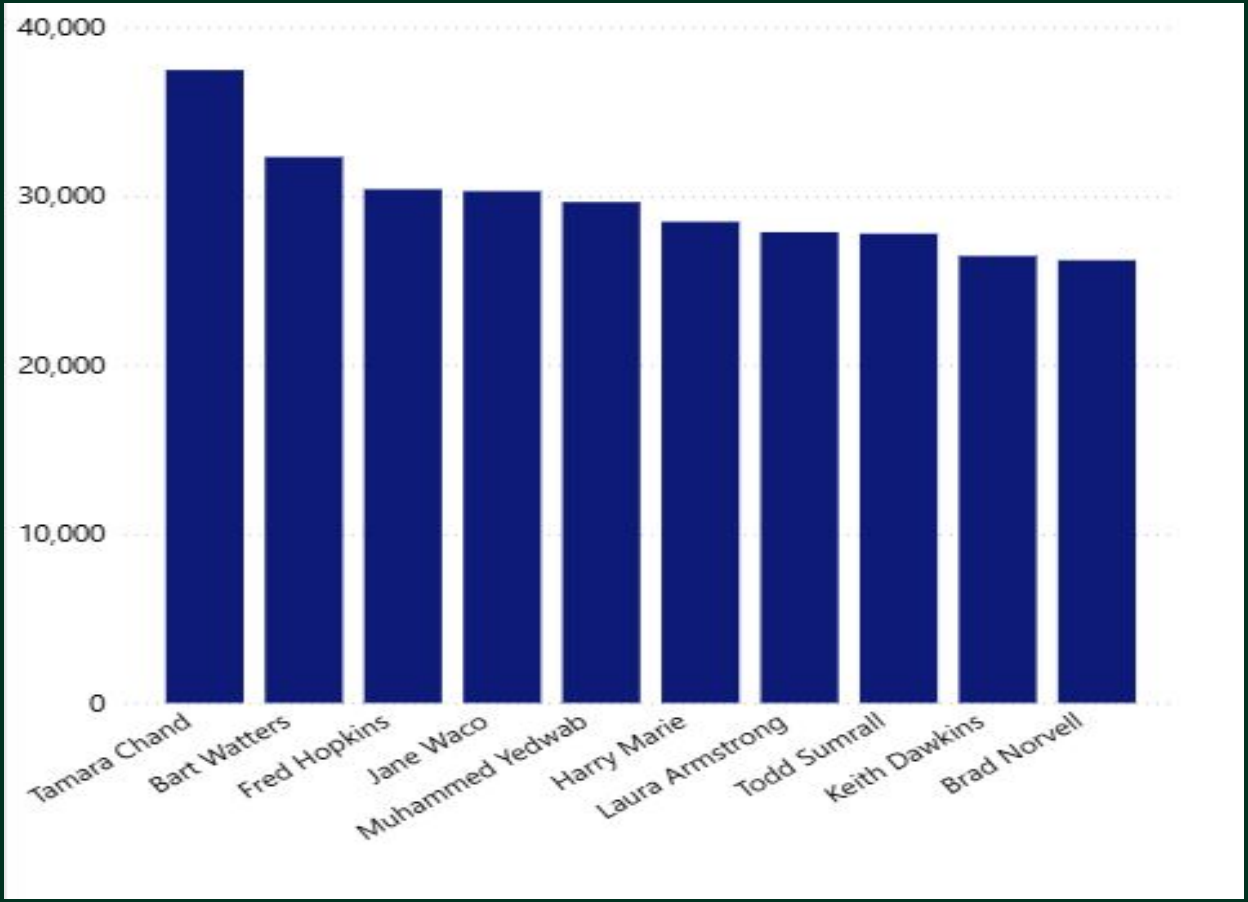
- Q12. Write a query to find the top 10 products by total sales
- Q13. Show yearly sales trends for all years.
- Q14. Show yearly sales trends for all months.
- Q15. Find customers who placed more than 30 orders in a single year.
- Q16. Compare sales vs. profit across different regions.
- Q17. Show monthly sales trends for the last 4 years.

Q1. Find out the total profit, sales, discount and shipping cost according to the segment.

```sql
(SELECT
    segment,
    ROUND(SUM(profit), 2) total_profit,
    ROUND(SUM(sales), 2) total_sales,
    ROUND(SUM(ShippingCost), 2) total_shippingcost,
    ROUND(SUM(discount), 2) total_discount
FROM
    orders
GROUP BY segment);
```

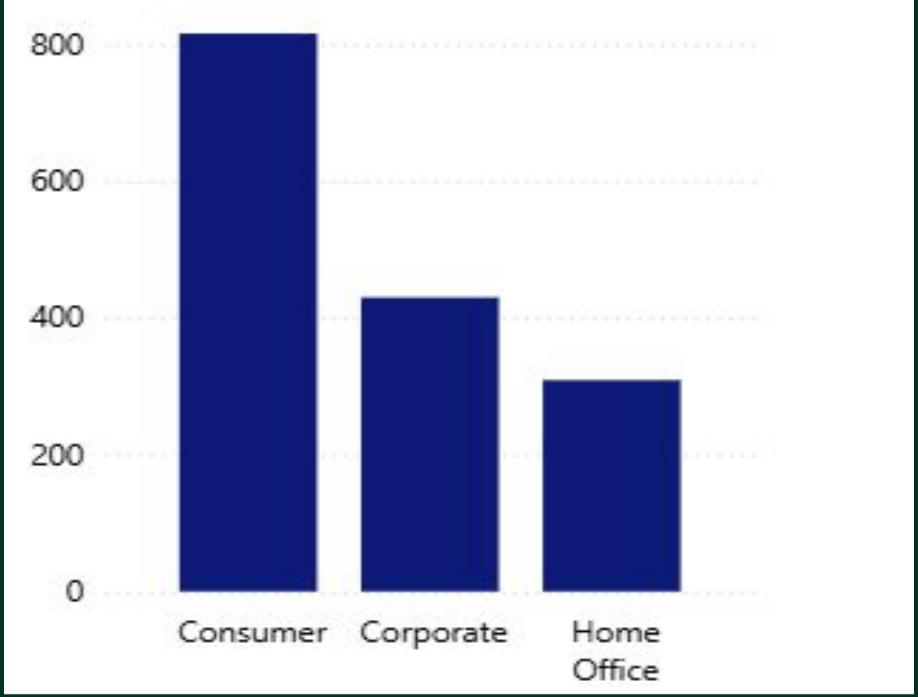
Q2. Top 10 corporate customers who gave the best sales.

```sql
select
    customername,round(sum(sales),2) AS Total_sales
from
    orders
where
    segment='corporate'
group by
    customername
order by
    total_sales desc
limit 10;
```


40,000
30,000
20,000
10,000
0
Tamara Chand
Bart Watters
Fred Hopkins
Jane Waco
Muhammed Yedwab
Harry Marie
Laura Armstrong
Todd Sumrall
Keith Dawkins
Brad Norvell

Q3. How many order received in India on each segment?

```sql
select
    segment,count(orderid) AS ordercount
from
    orders
where
    country="India"
group by
    segment;
```


800
600
400
200
0
Consumer
Corporate
Home Office

Q4. What would be the % of profit with respect to sales for ‘santiago’ state by each order priority.

```
(SELECT
    OrderPriority,
    ROUND((SUM(profit) / SUM(sales)) * 100, 2) AS Profit_Percentage
FROM
    orders
WHERE
    state = 'santiago'
GROUP BY OrderPriority);
```

Q5. What is the avg shipping cost of 1st class for top 15 customers in south-east asia for home office site?

```sql
(SELECT
    CustomerName, AVG(ShippingCost) AvgShippingCost
FROM
    orders
WHERE
    region = 'Southeast Asia'
        AND segment = 'Home Office'
        AND shipmode = 'First class'
GROUP BY CustomerName
ORDER BY AvgShippingCost DESC
LIMIT 15);
```

Q6. What are the total no of orders for each category by each segment order date wise?

```
(SELECT
    category, segment, YEAR(orderdate), COUNT(orderid)
FROM
    orders
GROUP BY category , segment , YEAR(orderdate));
```

Q7. How many product shipped by ‘first class’ in each priority.

```
(SELECT
    OrderPriority, COUNT(productid)
FROM
    orders
WHERE
    shipmode = 'First Class'
GROUP BY OrderPriority);
```

Q8 Find out total sales and profit for each category with low order priority and zero discount.

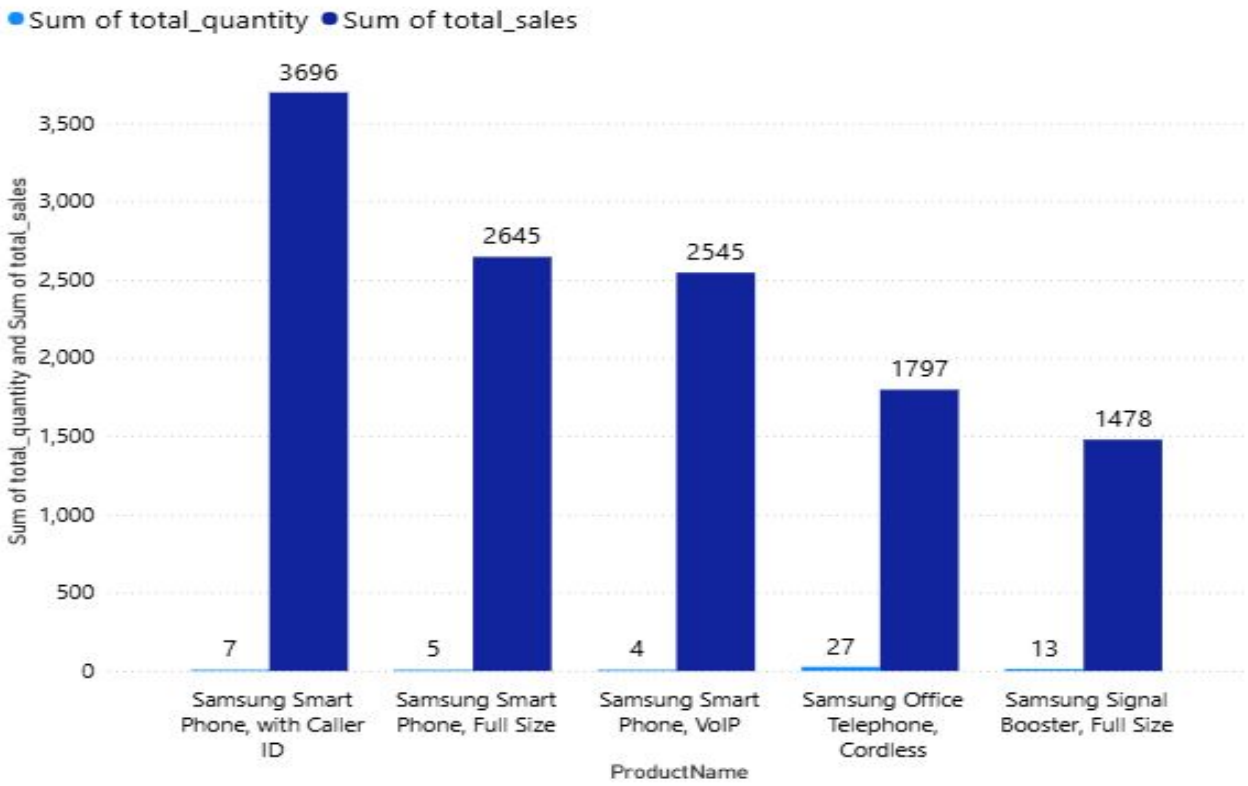
```
(SELECT
    category, SUM(sales) total_sales, SUM(profit) total_profit
FROM
    orders
WHERE
    Discount = 0
        AND OrderPriority RLIKE 'low'
GROUP BY category);
```

Q9. Name the cities of Australia with % of sales?

```
(SELECT
    city,
    CONCAT(ROUND((SUM(sales) / (SELECT
                                    SUM(Sales)
                                FROM
                                    orders
                                WHERE
                                    country = 'Australia')) * 100,
                        2),
            '%') AS Sales_Percentage
FROM
    orders
WHERE
    country = 'Australia'
GROUP BY city);
```

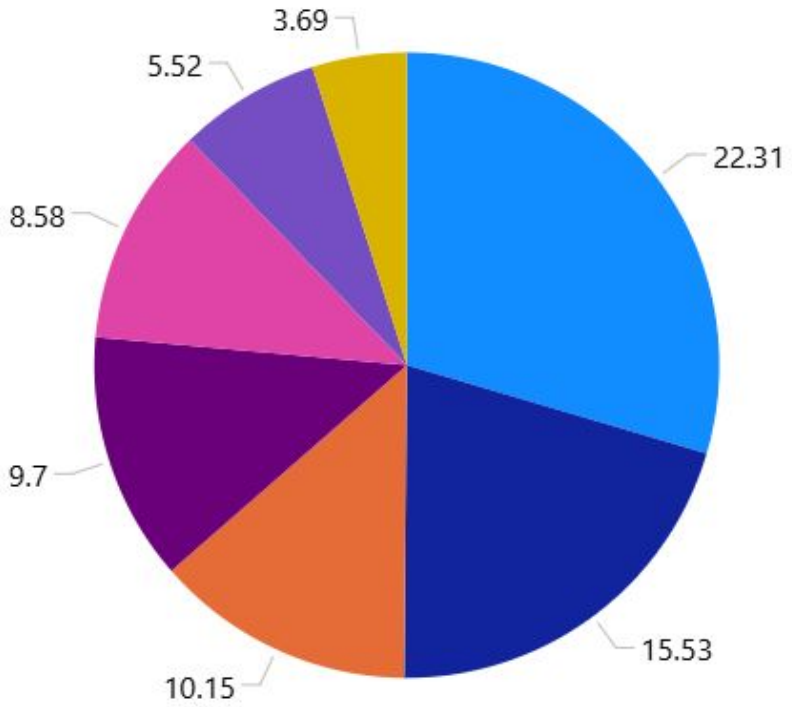
Q10.WHAT IS THE TOTAL SALES AND QUANTITY FOR TOP 5 SAMSUNG PRODUCTS IN APAC MARKET FOR THE YEAR 2014?

```sql
SELECT
    ProductName,
    SUM(Sales) AS total_sales,
    SUM(quantity) AS total_quantity
FROM
    orders
WHERE
    ProductName LIKE '%Samsung%'
        AND market = 'apac'
        AND YEAR(orderdate) = 2014
GROUP BY ProductName
ORDER BY total_sales
LIMIT 5;
```


Sum of total_quantity
Sum of total_sales
Sum of total_quantity and Sum of total_sales
3,500
3,000
2,500
2,000
1,500
1,000
500
0
3696
2645
2545
1797
1478
7
5
4
27
13
Samsung Smart Phone, with Caller ID
Samsung Smart Phone, Full Size
Samsung Smart Phone, VoIP
Samsung Office Telephone, Cordless
Samsung Signal Booster, Full Size
ProductName

Q11. What % of shipping cost contribute in sales for ‘Nokia’ Products in ‘NEW ZEALAND’ country?

```
CREATE VIEW Q11 AS
    (SELECT
        ProductName,
        ROUND((SUM(shippingcost) / SUM(sales)) * 100,
                2) shippingCost_Percent_outOfSales
    FROM
        orders
    WHERE
        ProductName LIKE '%nokia%'
            AND country = 'New Zealand'
    GROUP BY ProductName
    ORDER BY shippingCost_Percent_outOfSales DESC);
```



ProductName

- Nokia Signal Booster, VoIP
- Nokia Signal Booster, with Caller ID
- Nokia Office Telephone, VoIP
- Nokia Headset, VoIP
- Nokia Smart Phone, with Caller ID
- Nokia Smart Phone, Full Size
- Nokia Headset, Cordless

Q12.Write a query to find the top 10 products by total sales

```sql
SELECT
    productname, ROUND(SUM(sales), 2) AS Total_sales
FROM
    orders
GROUP BY productname
ORDER BY total_sales DESC
LIMIT 10;
```

Q13. Show yearly sales trends for all years

```sql
with cte as (SELECT
    EXTRACT(YEAR FROM orderdate) AS Yr,
    ROUND(SUM(Sales), 0) AS Current_total_sales
FROM
    orders
GROUP BY 1
ORDER BY 1),
cte2 as (select Yr,Current_total_sales,coalesce(lag(Current_total_sales,1) over(),"NA") as prev_total_sales
from cte)
SELECT
    Yr,
    Current_total_sales,
    prev_total_sales,
    COALESCE(CONCAT(ROUND(((Current_total_sales - prev_total_sales) / prev_total_sales) * 100,
                            2),
                    '%'),
            'NA') AS growth_percent
FROM
    cte2;
```

Q14 Show yearly sales trends for all months.

```sql
with cte as (SELECT
    EXTRACT(YEAR FROM orderdate) AS yr,
    EXTRACT(MONTH FROM orderdate) AS mnth,
    ROUND(SUM(sales), 2) AS current_mnth_total_sales
FROM
    orders
GROUP BY EXTRACT(YEAR FROM orderdate) , EXTRACT(MONTH FROM orderdate)
ORDER BY 1 , 2),
cte2 as (select yr,mnth,current_mnth_total_sales,
                coalesce(lag(current_mnth_total_sales,1)over(partition by yr),"NA") as prev_month_sales
         from cte)
SELECT
    yr,
    mnth,
    current_mnth_total_sales,
    prev_month_sales,
    COALESCE(CONCAT(ROUND(((current_mnth_total_sales - prev_month_sales) / prev_month_sales) * 100,
                          2),
                    '%'),
            'NA') AS growth_percent
FROM
    cte2;
```

Q15.Find customers who placed more than 30 orders in a single year.

```sql
(SELECT
    *
FROM
    (SELECT
        EXTRACT(YEAR FROM orderdate) yr,
            customerid,
            COUNT(orderid) total_orders
    FROM
        orders
    GROUP BY 1 , customerid
    ORDER BY 1 , 2) AS t1
WHERE
    total_orders > 30);
```

Q16. Compare sales vs. profit across different regions.

```sql
(SELECT
    region,
    ROUND(SUM(sales), 2) AS total_Sales,
    ROUND(SUM(profit), 0) AS total_profit
FROM
    orders
GROUP BY region);
```

Q17. Show monthly sales trends for the last 4 years.

```sql
with cte as (SELECT
    EXTRACT(MONTH FROM orderdate) AS mnth,
    EXTRACT(YEAR FROM orderdate) AS yr,
    ROUND(SUM(sales), 2) AS current_mnth_total_sales
FROM
    orders
GROUP BY 1 , 2
ORDER BY 1 , 2),
cte2 as (select yr,mnth,current_mnth_total_sales,
                coalesce(lag(current_mnth_total_sales,1)over(partition by mnth),"NA") as prev_month_sales
        from cte)
SELECT
    mnth,yr,
    current_mnth_total_sales,
    prev_month_sales,
    COALESCE(CONCAT(ROUND(((current_mnth_total_sales - prev_month_sales) / prev_month_sales) * 100,
                            2),
                    '%'),
            'NA') AS growth_percent
FROM
    cte2
ORDER BY 1 , 2;
```

Thank You